

Case Study 2: A Statistical Analysis of Suspected Pay Inequality Based on Race and Region.

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2025-04-09

For decades, wage disparities in the United States have posed significant economic challenges for different minority groups whether it be based on race, gender, or education. The most defining and enduring form of pay inequality has existed between Black and White male workers in the U.S. Even if both groups have the same years of professional experience or educational background there may still be a wage gap present between their weekly salaries. The U.S. Census Bureau is interested in investigating if the wage gap between these two groups changes depending on the region in which they are working. To dive into this question we will be utilizing the March 1988 U.S. Current Population Survey which has the economic status and activities of the American populace. However, for our investigation we will be looking at the economic status breakdown of a sample of 25,632 male respondents. We will then fit a model to identify if in 1988 Black workers were paid significantly less than White workers by holding education and experience constant. This will also inform us if the pay gap varies across different regions.

Data Description and Exploratory Data Analysis

For our linear regression modeling, we will be utilizing the Sleuth2 library that contains the dataset for the March 1988 CPS sample of 25,632 male respondents. This dataset contains information on whether a given male respondent was black or white (not including other races), years of work experience, years of education, region in which they are working, weekly wages from the previous year (measured in 1992 dollars), and proximity to a metropolitan area. However, we will only be using education, wage, race, experience, and region. As previously mentioned this economic status information was gathered from the U.S. March 1988 CPS survey. Additionally, to determine whether a respondent is Black or White their response of yes or no to being Black are converted to 1 or 0.

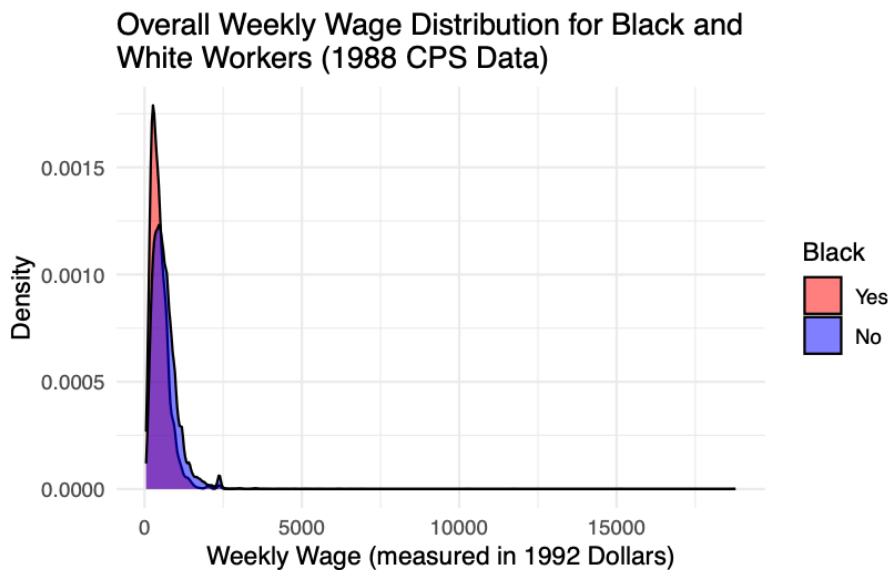
Our research question primarily focuses on finding if the wage gap between Black and White workers varied between regions even if they have the same years of work experience and

education. Due to this we removed the variable SMSA or proximity for a metropolitan area from our model.

First, we will analyze the overall wage gap differences between Black and White workers. To do this we will create a table for the median, standard deviation, minimum, and max weekly wage as well as a distribution graph.

Table 1: Overall Weekly Wage Summary for Black and White Workers (1988 CPS Data)

Black	Median	SD	Min	Max
Yes	412.29	307.81	53.83	3527.34
No	569.80	451.27	50.39	18777.20



It should be noted that the density plot includes the extreme max values seen in the overall weekly wage table. Due to these extreme values the graph is slightly distorted. However, it can still be assessed that White workers, indicated by the blue distribution, skew more to the right indicating a higher overall weekly wage in 1988. On the other hand, Black workers skew to the left with a high peak indicating a larger concentration in the lower end of overall weekly wage distributions.

From this exploration there does seem to be evidence to suspect potential wage differences between Black and White workers in their overall weekly wage distribution. However, what about their regional wage gap difference? To further investigate this we will apply statistical modeling and analysis.

Statistical Analysis & Results

When fitting our model we found that our original data did not perfectly meet the conditions for normality and equal variance. It did meet linearity via the residuals vs fitted plot and independence as it was a large scale national survey with randomly sampled responses from Black and White male workers. To improve our models normality and equal variance we attempted to $\log()$ transform it. After the transformation, the tail deviations in the normality plot seemed more pronounced. We are assuming it is caused by the extreme wage values found in the 25,631 randomly selected sample population. Due to the transformation not improving the conditions we chose to work with the original data.

Since we want to discover if region has an effect on weekly wage based on an interaction with the worker's race we will create two models. The reduced model will show no interaction between race and region while the interactive model will.

First we will fit a reduced model where region does not have an effect on the weekly wage for Black or White workers. In this model we will also include education and experience as it is being assumed that workers with the same years of experience in both categories would still experience a wage gap based on race. The model can be described as:

$$E[Wage|Black, Region] = \beta_0 + \beta_1(BlackNo) + \beta_2(RegionNE) + \beta_3(RegionS) + \beta_4(RegionW) + \beta_5(Education) + \beta_6(Experience)$$

Wage denotes the weekly wage received by an individual male respondent. BlackNo indicates whether or not an individual is Black. The different regions described by β_2 to β_4 indicate the region where the individual works. If all of those betas are 0 it means they work in the Mid West. Education and Experience indicate the length in years they have in those categories which would influence wage. The following table indicates the intercepts of the model:

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	-488.90	16.49	-29.64	0.00
BlackNo	118.91	9.44	12.60	0.00
RegionNE	35.43	7.20	4.92	0.00
RegionS	-22.44	6.78	-3.31	0.00
RegionW	20.10	7.37	2.73	0.01
Education	62.29	0.89	69.60	0.00
Experience	10.73	0.21	51.46	0.00

Next, we will fit an interaction model where region does have an effect on weekly wage based on race. The model can be described as:

$$\begin{aligned}
E[Wage|Black, Region] = & \beta_0 + \beta_1(BlackNo) \\
& + \beta_2(RegionNE) + \beta_3(RegionS) + \beta_4(RegionW) \\
& + \beta_5(Education) + \beta_6(Experience) + \beta_7(BlackNo)(RegionNE) \\
& + \beta_8(BlackNo)(RegionS) + \beta_9(BlackNo)(RegionW)
\end{aligned}$$

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	-475.14	25.65	-18.52	0.00
BlackNo	104.62	22.74	4.60	0.00
RegionNE	25.87	31.16	0.83	0.41
RegionS	-42.97	25.01	-1.72	0.09
RegionW	20.52	37.38	0.55	0.58
Education	62.28	0.90	69.56	0.00
Experience	10.72	0.21	51.44	0.00
BlackNo:RegionNE	10.07	32.03	0.31	0.75
BlackNo:RegionS	22.46	25.99	0.86	0.39
BlackNo:RegionW	-0.15	38.13	0.00	1.00

To determine which model would be a better fit for the data and if the interaction between race and region has a statistically significant effect on a worker's weekly wage we will perform an F-test. This test compares the interactive model with the reduced model to determine appropriate fit.

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
25621	4042067674	NA	NA	NA	NA
25624	4042236918	-3	-169243.7	0.3575887	0.7836525

After conducting an F-test comparing both models the test yielded $F = 0.357$ and a p-value of 0.7836525 which is larger than $\alpha = 0.05$. This indicates that there is no statistically significant relationship between race and region for us to justify using the interactive model. Therefore, we can continue the analysis with the reduced model to determine the overall pay gap estimate between Black and White workers in each region.

To calculate the wage gap differences we will be using the following reduced model:

$$\begin{aligned}
\widehat{Wage} = & -488.90 + 118.91(BlackNo) \\
& + 35.43(RegionNE) - 22.44(RegionS) + 20.10(RegionW) \\
& + 62.29(Education) + 10.73(Experience)
\end{aligned}$$

Since we are holding Experience and Education constant we will find the mean quantity of both categories so that we can accurately evaluate the wage gap. From the calculations we know

that the mean Experience is 18.58 years and Education has a mean of 13.07 years. Once we evaluated our model for each region and race of the workers we found that across all regions that wage gap was \$118.91 (measured in 1992 dollars). This predicted wage gap between White and Black workers is supported by the resulting confidence interval where we are 95% confident that the true wage gap between White and Black workers lies between \$100.41 and \$137.41. Our predicted wage gap of \$118.91 falls in this interval.

Discussion

When we began this study our goal was to determine if the suspected wage gap between Black and White workers varied between regions and evaluate the wage gaps. We first compared a reduced model where there is no interaction between race and region with an interactive model where race and region would have an influence on the predicted wage. After evaluating them we found that there was no statistically significant relationship between race and region to justify using the interactive model. This meant that there was no statistically significant variation in wage gaps in different regions based on race. Ultimately, the wage gap that existed between White and Black workers in 1988 was \$118.91 (measured in 1992 dollars) when holding education and experience constant using their mean values. Our predicted wage gap is also supported by our confidence interval where we are 95% confident that the true wage gap between White and Black workers lies between \$100.41 and \$137.41. However, our study does pose limitations as information from 1988 is outdated and may not be useful in a 2025 context. Additionally, we did not use a transformation for our data which has extreme wage values. In future studies, creating a better model fit using transformations could help inform the statistical model.

R Appendix

```
library(knitr)
library(broom)
library(dplyr)
library(kableExtra)
library(ggplot2)
library(stringr)
# Loading the Statistical Sleuth data library
library(Sleuth2)
# Reading in and saving the data
wages <- Sleuth2::ex1029
```

```

#Factoring
wages$Region <- factor(wages$Region)
wages$Black <- factor(wages$Black)

#Summary statistics of the wages for Black and White Workers overall
wage_summary <- wages |>
  group_by(Black) |>
  summarize(
    Median_Wage = round(median(Wage),2),
    SD_Wage = round(sd(Wage), 2),
    Min_Wage = round(min(Wage),2),
    Max_Wage = round(max(Wage),2),
    .groups = "drop"
  )

#Creating the summary table
wage_summary |>
  kbl(col.names = c("Black",
                    "Median",
                    "SD",
                    "Min",
                    "Max"),
      caption = "Overall Weekly Wage Summary for Black and White Workers (1988 CPS Data)",
      align = "c") |>
  kable_classic(full_width = FALSE, latex_options = c("HOLD_position"))

```

Table 2: Overall Weekly Wage Summary for Black and White Workers (1988 CPS Data)

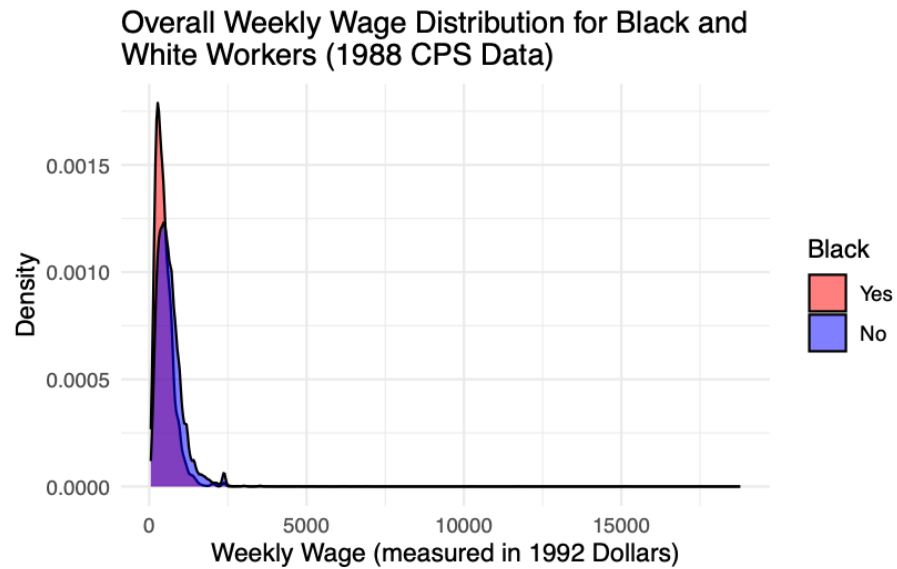
Black	Median	SD	Min	Max
Yes	412.29	307.81	53.83	3527.34
No	569.80	451.27	50.39	18777.20

```

#Plot for the distribution chart
ggplot(wages, aes(x = Wage, fill = Black)) +
  geom_density(alpha = 0.5) +
  labs(
    title = str_wrap("Overall Weekly Wage Distribution for Black and White Workers (1988 C
    x = "Weekly Wage (measured in 1992 Dollars)",
    y = "Density"
  ) +

```

```
scale_fill_manual(values = c("No" = "blue", "Yes" = "red"), name = "Black") +
theme_minimal()
```



```
#Summary statistics of the wages for Black and White Workers in different regions.
wage_summary <- wages |>
  group_by(Region, Black) |>
  summarize(
    Median_Wage = round(median(Wage),2),
    SD_Wage = round(sd(Wage), 2),
    Min_Wage = round(min(Wage),2),
    Max_Wage = round(max(Wage),2),
    .groups = "drop"
  )

#Creating the summary table
wage_summary |>
  kbl(col.names = c("Region",
                    "Black",
                    "Median",
                    "SD",
                    "Min",
```

```

      "Max"),
    caption = "Weekly Wage Summary by Region for Black and White Workers (1988 CPS Data)",
    align = "c") |>
kable_classic(full_width = FALSE, latex_options = c("HOLD_position"))

```

Table 3: Weekly Wage Summary by Region for Black and White Workers (1988 CPS Data)

Region	Black	Median	SD	Min	Max
MW	Yes	474.83	304.46	59.67	2374.15
MW	No	593.00	446.40	54.61	15123.50
NE	Yes	471.98	342.59	59.35	2374.15
NE	No	617.28	430.80	54.25	6913.58
S	Yes	375.74	287.30	61.73	3527.34
S	No	522.32	470.70	50.39	18777.20
W	Yes	493.83	337.45	53.83	2374.15
W	No	569.80	447.99	51.44	10288.00

This table was not ultimately used in the paper, but it was interesting to share. It should be noted that this summary table does not take into account education and experience levels as it is just a summary of overall weekly wage differences by region for Black and White workers.

```

#Creating the non-interaction model
no_interaction_md <- lm(Wage ~ Black + Region + Education + Experience, data = wages)

#Creating the summary table
no_interaction_table <- no_interaction_md |> tidy()

no_interaction_table |>
  kbl(col.names = c("Coefficients",
                    "Estimates",
                    "Standard Error",
                    "t-value",
                    "p-value"),
      align = "c", booktabs = T, linesep="",
      digits = c(2, 2, 2, 2)) |>
  kable_classic(full_width = FALSE, latex_options = c("HOLD_position"))

```


Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	-488.90	16.49	-29.64	0.00
BlackNo	118.91	9.44	12.60	0.00
RegionNE	35.43	7.20	4.92	0.00
RegionS	-22.44	6.78	-3.31	0.00
RegionW	20.10	7.37	2.73	0.01
Education	62.29	0.89	69.60	0.00
Experience	10.73	0.21	51.46	0.00

```
#Creating the non-interaction model
no_interaction_md <- lm(Wage ~ Black+Region + Education + Experience, data = wages)

#Summary of the non-interaction model
summary(no_interaction_md)
```

Call:

```
lm(formula = Wage ~ Black + Region + Education + Experience,
    data = wages)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-1098.7  -210.3   -50.0   143.3  18140.8
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -488.8982    16.4929  -29.643  < 2e-16 ***
BlackNo      118.9079     9.4393   12.597  < 2e-16 ***
RegionNE     35.4276     7.2027    4.919 8.77e-07 ***
RegionS     -22.4357     6.7788   -3.310 0.000935 ***
RegionW      20.0999     7.3678    2.728 0.006374 **
Education    62.2900     0.8949   69.605  < 2e-16 ***
Experience    10.7250     0.2084   51.461  < 2e-16 ***
---
```

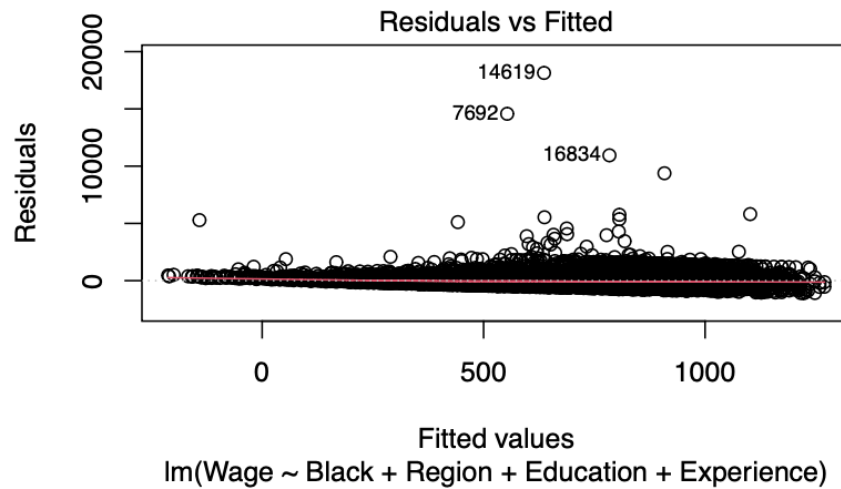
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 397.2 on 25624 degrees of freedom

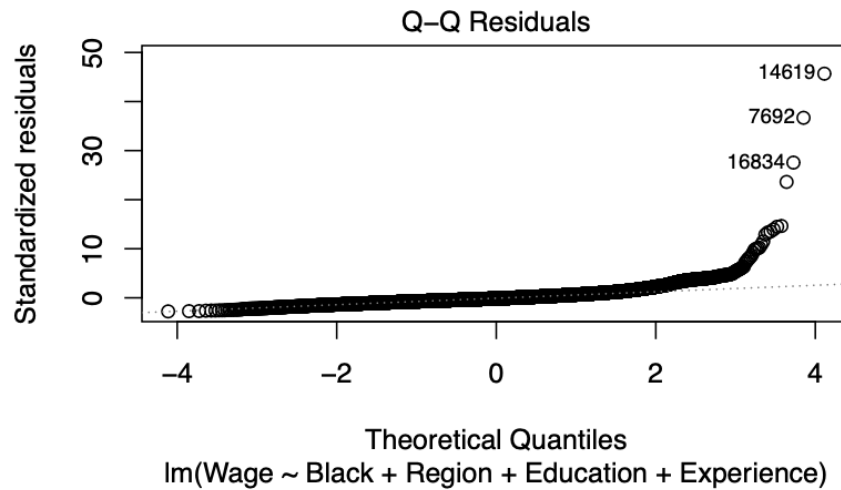
Multiple R-squared: 0.201, Adjusted R-squared: 0.2008

F-statistic: 1074 on 6 and 25624 DF, p-value: < 2.2e-16

```
#Res vs Fitted  
plot(no_interaction_md, which = 1)
```

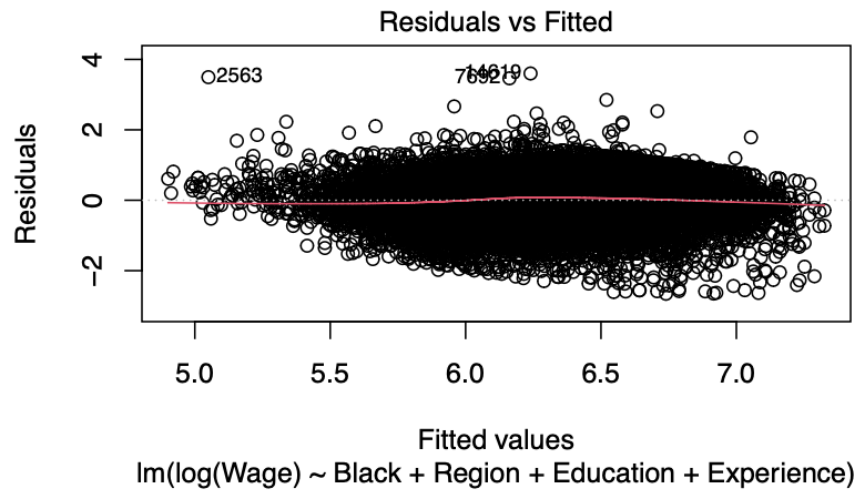


```
#Q-Q Plot  
plot(no_interaction_md, which = 2)
```

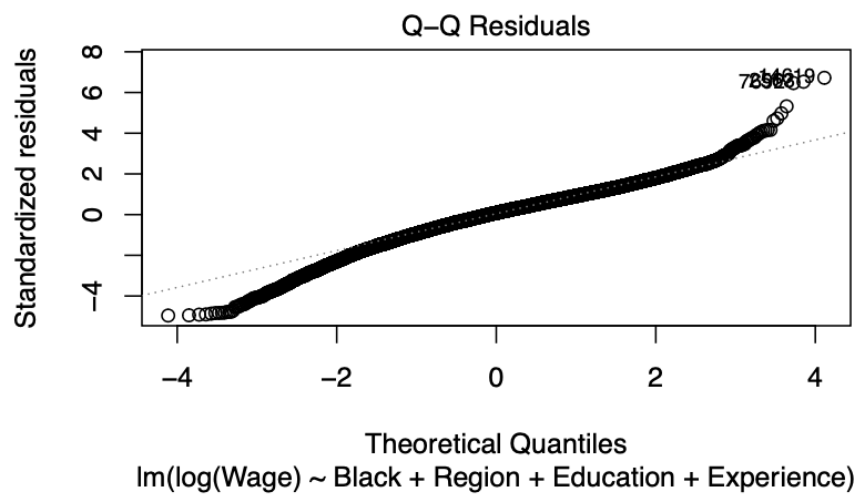


```
#Checking the log() transformation of the non-interaction model
log_no_interaction_md <- lm(log(Wage) ~ Black+Region + Education + Experience, data = wage)

#Res vs Fitted
plot(log_no_interaction_md, which = 1)
```



```
#Q-Q Plot  
plot(log_no_interaction_md, which = 2)
```



```
#Creating the interaction model
interaction_md <- lm(Wage ~ Black*Region + Education + Experience, data = wages)

#Creating the summary table
interaction_table <- interaction_md |> tidy()

interaction_table |>
  kbl(col.names = c("Coefficients",
                    "Estimates",
                    "Standard Error",
                    "t-value",
                    "p-value"),
      align = "c", booktabs = T, linesep="",
      digits = c(2, 2, 2, 2)) |>
  kable_classic(full_width = FALSE, latex_options = c("HOLD_position"))
```

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	-475.14	25.65	-18.52	0.00
BlackNo	104.62	22.74	4.60	0.00
RegionNE	25.87	31.16	0.83	0.41
RegionS	-42.97	25.01	-1.72	0.09
RegionW	20.52	37.38	0.55	0.58
Education	62.28	0.90	69.56	0.00
Experience	10.72	0.21	51.44	0.00
BlackNo:RegionNE	10.07	32.03	0.31	0.75
BlackNo:RegionS	22.46	25.99	0.86	0.39
BlackNo:RegionW	-0.15	38.13	0.00	1.00

```
#Creating the interaction model
interaction_md <- lm(Wage ~ Black*Region + Education + Experience, data = wages)

#Summary of the interaction model
summary(interaction_md)
```

Call:

```
lm(formula = Wage ~ Black * Region + Education + Experience,
    data = wages)
```

Residuals:

```

      Min       1Q   Median       3Q      Max
-1099.7  -210.2   -49.8   143.5 18139.6

```

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -475.1390    25.6513  -18.523  < 2e-16 ***
BlackNo       104.6164    22.7401   4.601  4.23e-06 ***
RegionNE      25.8664    31.1619   0.830   0.4065
RegionS     -42.9668    25.0128  -1.718   0.0858 .
RegionW       20.5236    37.3788   0.549   0.5830
Education     62.2778     0.8953  69.561  < 2e-16 ***
Experience    10.7225     0.2084  51.440  < 2e-16 ***
BlackNo:RegionNE 10.0695    32.0283   0.314   0.7532
BlackNo:RegionS  22.4551    25.9883   0.864   0.3876
BlackNo:RegionW  -0.1473    38.1314  -0.004   0.9969
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 397.2 on 25621 degrees of freedom
 Multiple R-squared: 0.201, Adjusted R-squared: 0.2007
 F-statistic: 716.2 on 9 and 25621 DF, p-value: < 2.2e-16

```

#F-statistic comparison of both models
anova(interaction_md, no_interaction_md)

```

Analysis of Variance Table

```

Model 1: Wage ~ Black * Region + Education + Experience
Model 2: Wage ~ Black + Region + Education + Experience
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1  25621 4042067674
2  25624 4042236918 -3    -169244 0.3576 0.7837

```

```
kable(anova_models)
```

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
25621	4042067674	NA	NA	NA	NA
25624	4042236918	-3	-169243.7	0.3575887	0.7836525

```
#Finding the mean of Experience and Education to hold them constant in the calculation for  
edu_mean <- mean(wages$Education)  
exp_mean <- mean(wages$Experience)
```

```
edu_mean
```

```
[1] 13.07627
```

```
exp_mean
```

```
[1] 18.58656
```

```
#I did the wage gap calculations by hand.
```

```
#Calculating the confidence interval, but we are only interested in BlackNo  
confint(no_interaction_md)
```

	2.5 %	97.5 %
(Intercept)	-521.225242	-456.57108
BlackNo	100.406412	137.40948
RegionNE	21.309886	49.54535
RegionS	-35.722546	-9.14885
RegionW	5.658736	34.54116
Education	60.535891	64.04405
Experience	10.316516	11.13351